

STRONG+

Extended Karplus-Strong Synthesis

Archive — The History, People, Algorithm and Culture of KS Synthesis

This document is the companion archive to the Strong+ browser synthesizer — an implementation of the Extended Karplus-Strong (EKS) algorithm with commuted synthesis body resonance. Strong+ was built as a vibe coding project by Yuwa (Positive Lofi) in dialogue with Claude (Anthropic), 2025.

The Karplus-Strong algorithm is public domain (Karplus & Strong, Computer Music Journal, MIT Press, 1983). All historical information is sourced from published academic papers, CCRMA documentation, and verified public records.

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1. The Story

An Accident That Sounded Like a String

The story begins not with a musical vision but with boredom. Around 1978-79, **Alex Strong** — a computer science student at Stanford who also played viola in a string quartet — was experimenting with wavetable synthesis on an 8-bit microprocessor. Wavetable synthesis was simple: store a waveform in a buffer, loop it. It was also, as Karplus and Strong later wrote, "rather boring, as any precisely periodic sound tends to be."

Strong had a small idea. Instead of reading the wavetable unchanged, filter it while reading — average each sample with its neighbor. A two-point average. No multiplications needed, just one addition and a shift. The cheapest possible filter on the cheapest possible hardware. The result sounded like a plucked string. Not a simulation. Something that felt genuinely alive — with a natural, physical decay, with harmonics that faded in the right order. "Almost no matter what initial table was used," the paper would say. Even random noise as input sounded like a string.

This was the accident. Strong brought it to Karplus, who did the math. Kevin Karplus analyzed why it worked using Z-transforms. They built hardware — eventually a custom VLSI chip they named "**Digital**" (digital + guitar) — and published in 1983. Neither had set out to synthesize a string. They had set out to make wavetable synthesis less boring.

The Composer Who Made It Music

David A. Jaffe was a composition PhD student at CCRMA — Stanford's Center for Computer Research in Music and Acoustics. He had grown up playing violin, mandolin, and guitar in bluegrass and folk bands before turning to composition. He played in the same string quartet as Alex Strong.

When Strong showed him the algorithm, Jaffe recognized something the computer scientists hadn't fully articulated: this was a compositional instrument, not just a DSP trick. He brought it to **Julius O. Smith III**, who was finishing his PhD thesis on digital filter design applied to the violin. Smith looked at the transfer function of the KS loop and recognized something extraordinary: it coincided exactly with the transfer function of an idealized vibrating string. Karplus and Strong had suspected this — they had tried to prove it and failed. Smith demonstrated that the KS transfer function corresponded to an idealized vibrating string model. The algorithm wasn't approximating string behavior. It was a string model, derived accidentally from a completely different direction.

Together Jaffe and Smith developed the **Extended Karplus-Strong (EKS) algorithm** — adding six musical controls: tuning precision (H-eta), pick position (H-beta), dynamic level (H-L), pick direction (H-p), stiffness/dispersion (H-s), and explicit loop gain (H-d). They published in 1983 in the same issue of *Computer Music Journal* as the original KS paper — two papers, one discovery, two very different perspectives.

Silicon Valley Breakdown — A Symphony of Imaginary Instruments

Jaffe's piece **Silicon Valley Breakdown** (1982, premiered Venice Biennale 1983) is the artistic monument of this discovery. An orchestra of **imaginary plucked string instruments** — from a "piccolo mandolin" to an immense bass "plucked Golden Gate Bridge." Four ensembles in quadraphonic space, each with its own tone quality and position. Premiered at the Venice Biennale 1983; performed in 28+ countries. Hailed by *Le Monde* as a landmark of computer music.

The key insight: **the instrument and the room are not separate things**. Jaffe described the technique as blurring "the dividing line between string resonance and reverberation, between instrument and space." This is not a philosophical position — it is a technical one, and it would eventually become the theoretical basis for commuted synthesis.

The NeXT Machine — Two Revolutions, One Black Cube

Steve Jobs personally hired Jaffe and Smith (1986-91) to build the **NeXT Music Kit** — the first architecture to unify the Music-N and MIDI paradigms. First demonstrated at the October 12, 1988 NeXT launch at Davies Symphony Hall, San Francisco — the presentation ended with a live duet between the Cube and a violinist from the SF Symphony.

Independently, **Tim Berners-Lee** began building the first web browser and server at CERN in September 1990, on a NeXT Cube. The browser (WorldWideWeb.app) was written in Objective-C on NeXTSTEP — the same language, the same OS as the Music Kit. The first website went live December 25, 1990. Then, within twelve months: the Music Kit was open-sourced to CCRMA (1992), and CERN put the World Wide Web in the public domain on **April 30, 1993**. Two teams, one machine, two acts of radical openness.

2. The People

Alex Strong

INVENTOR · VIOLIST · BLUEGRASS MUSICIAN

A CS student at Stanford and practicing violist, Strong invented the algorithm c.1978-79 while experimenting with wavetable modification on an 8-bit microprocessor. The two-point average was designed to filter the wavetable on each reading pass — the string sound was a complete surprise. He co-designed the "Digitar" VLSI chip supporting 16 polyphonic voices. Later published a superior method as a patent (never commercialized).

Kevin Karplus

ANALYST · COMPUTER SCIENTIST · STANFORD

Karplus performed the first mathematical analysis of what Strong had discovered intuitively, characterizing the algorithm's frequency response using Z-transforms. Co-designed the Digitar VLSI chip with Strong. Their 1983 paper in *Computer Music Journal* (MIT Press, vol.7 no.2) established the algorithm formally. Karplus later moved to bioinformatics at UC Santa Cruz.

David A. Jaffe

COMPOSER · INNOVATOR · CCRMA STANFORD

Born April 29, 1955. Composition PhD student at CCRMA who played in the same string quartet as Alex Strong. Gave the algorithm its first musical voice in *May All Your Children Be Acrobats* (1981). Created *Silicon Valley Breakdown* (1982) — the landmark work. With Julius Smith developed the EKS algorithm. Created the NeXT Music Kit for Steve Jobs (1986-91). Has composed over 90 pieces for orchestra, chorus, chamber and electronic ensembles.

Julius O. Smith III

THEORIST · DSP ENGINEER · PROFESSOR EMERITUS CCRMA

Smith recognized that the KS transfer function coincided with that of an ideal vibrating string — proving what Karplus and Strong had conjectured. Derived KS as a special case of digital waveguide synthesis, opening a unified theoretical framework for physical modeling of strings, tubes, membranes, and vocal tracts. 1983 PhD thesis at Stanford applied these ideas to violin synthesis. Later co-invented the core audio fingerprinting algorithm behind Shazam. His CCRMA textbooks (pasp.html) remain the definitive reference for physical audio signal processing.

3. Timeline 1978–2025

1978-79	Alex Strong discovers the algorithm on an 8-bit microprocessor at Stanford. A two-point average on the wavetable produces unexpectedly convincing string sounds.
1981	May All Your Children Be Acrobats, Jaffe. First musical use of KS — 8 guitars, mezzo-soprano, computer tape. Text from Carl Sandburg's The People, Yes.
1982	Jaffe and Smith present the Extended KS (EKS) algorithm at ICMC Venice. Silicon Valley Breakdown composed.
Oct 1983	Two papers published simultaneously in Computer Music Journal (MIT Press, vol.7 no.2): the original KS paper and the EKS extensions. Smith's PhD thesis links KS to digital waveguide synthesis. Patent licensed to a company that promptly goes bankrupt.
1984	Telegram to the President, Jaffe — string quartet + tape. First major mixed acoustic/electronic KS work.
Venice 1983	Silicon Valley Breakdown premieres at the Venice Biennale. Hailed by Le Monde as a landmark of computer music.
1986-91	Steve Jobs recruits Jaffe and Smith for the NeXT Music Kit. First demonstrated October 12, 1988 at Davies Symphony Hall with a live duo between the Cube and an SF Symphony violinist.
1987	Grass, Jaffe — SSA female chorus + tape (Carl Sandburg). Uses Smith's waveguide time-varying reverb networks — early commuted synthesis.
Sep 1990	Tim Berners-Lee begins building WorldWideWeb.app on a NeXT Cube at CERN.
Dec 1990	First website goes live on a NeXT Cube at CERN.
1989	Stanford signs agreement with Yamaha (Sondius patent package) to develop digital waveguide synthesis commercially.
1992	NeXT Music Kit open-sourced to CCRMA. Smith formalizes commuted synthesis.
Apr 30, 1993	CERN puts the World Wide Web in the public domain. Two revolutions, one machine, both open-sourced within 12 months.
1994	Yamaha VL1 — first commercial physical modeling synthesizer. 2-voice, extraordinarily expressive. Direct descendant of KS via waveguide synthesis.
~2010	All original Karplus-Strong and waveguide synthesis patents expire. The algorithm is fully open.
2015	Mutable Instruments Rings — brings KS-adjacent string synthesis to Eurorack. One of the most celebrated modules ever made.
2025	Strymon Superkar+ — 32-voice Eurorack module, described as the one of the most advanced dedicated hardware KS instruments currently available.

4. Musical Works

1981

May All Your Children Be Acrobats

David A. Jaffe · 8 guitars, mezzo-soprano, computer tape · ~15'

The first musical use of the KS algorithm. Jaffe had planned FM synthesis for the guitar sounds but Strong's method proved far more convincing. Text from Carl Sandburg's *The People, Yes* — a celebration of human resilience. The synthetic strings blend with the mezzo-soprano, blurring the line between acoustic and computer-generated for the first time.

[first musical use of KS] [Sandburg text] [premiered 1981]

1982 / premiered Venice 1983

Silicon Valley Breakdown

David A. Jaffe · computer-generated plucked strings · quadraphonic · ~18'

The landmark work. An orchestra of imaginary plucked string instruments — from a "piccolo mandolin" to an immense bass "plucked Golden Gate Bridge." Four ensembles in quadraphonic space. Premiered Venice Biennale 1983. Hailed by *Le Monde*. Performed in 28+ countries. Synthesized on the Systems Concept Digital Synthesizer at CCRMA. Described by Jaffe as blurring "the dividing line between string resonance and reverberation, between instrument and space."

[EKS algorithm] [quadraphonic] [Venice Biennale 1983] [landmark]

1984

Telegram to the President

David A. Jaffe · string quartet + computer tape · ~12'

A political work for live string quartet and KS-synthesized tape. The algorithm creates synthetic doubles of the acoustic quartet — at times indistinguishable. First major mixed acoustic/electronic KS work.

[mixed acoustic/electronic] [EKS] [1984]

1987

Grass

David A. Jaffe · SSA female chorus + computer tape · 11'30"

Based on Carl Sandburg's poem about the healing of war's wounds. The tape uses KS string synthesis and Julius Smith's waveguide time-varying reverberation networks — an early form of commuted synthesis. The string sounds are airy, vocal, almost indistinguishable from human voices. Commissioned for the Skidmore College Chorus.

[waveguide reverb] [early commuted synthesis] [Sandburg] [1987]

5. The Algorithm — KS and EKS

The Original Algorithm (Karplus & Strong 1983)

Initialize a circular buffer (delay line) of N samples with white noise. On each cycle, replace each sample with the average of the current and previous sample and write it back. That is the entire Karplus-Strong algorithm.

$$y[n] = 0.5 * (y[n-N] + y[n-N-1])$$

The two-point average is a one-zero lowpass filter $H(z) = 0.5 + 0.5z^{-1}$. It attenuates high frequencies more than low ones — the string loses its upper harmonics progressively, exactly as a real string does. N samples = one period: pitch = sampleRate / N. The algorithm requires no multiplications — designed for 8-bit microprocessors.

The Extended KS Algorithm (Jaffe & Smith 1983)

Six filters, each solving a specific musical problem:

H_p(z) — Pick Direction LP

$$(1-p) / (1 - p*z^{-1})$$

Controls pick direction. p=0: downstroke. p>1: opposite. Shapes the excitation timbre before it enters the loop.

H_beta(z) — Pick Position Comb

$$1 - z^{-(\text{floor}(\beta * N))}$$

Simulates pick position. Plucking near the bridge cancels harmonics at that node, producing a nasal, muted timbre.

H_L(z) — Dynamic Level LP

$$(1-R_L) / (1 - R_L*z^{-1})$$

Maps velocity to excitation bandwidth. Loud=bright. Soft=dark. Physically accurate to how strings respond to impact force.

H_eta(z) — Fractional Delay Tuning

$$-(\eta - z^{-1}) / (1 - \eta*z^{-1})$$

Solves the integer-N tuning problem. A first-order allpass introduces a fractional delay to compensate pitch error when N must be rounded.

H_s(z) — Stiffness / Dispersion

$$\text{Allpass chain}$$

Models string stiffness: real strings have slightly stretched harmonics (inharmonic). In practice a very subtle correction — keep values low to avoid artifacts.

H_d(z) — Loop Gain / Decay

$$g * H_{\text{material}}(z)$$

Controls decay rate. g<1 means energy is lost per cycle. Combined with the one-zero LP filter, shapes both sustain and timbre evolution over time.

Commutated Synthesis (Smith 1992)

Since both the string and the instrument body are linear time-invariant (LTI) systems, their order in the signal chain can be commuted (swapped) without changing the output. The body impulse response is folded into the excitation signal — the string plucks a pre-convolved body. This eliminates hundreds of resonator filter sections while achieving the same result. In Strong+, the BODY section applies an IR before the string loop — the reverb is not an effect; it is the instrument body.

From KS to Digital Waveguide Synthesis

Smith showed that KS is a degenerate case of a bidirectional digital waveguide: two delay lines propagating traveling waves in opposite directions. The full theory models strings, tubes, membranes, and vocal tracts — any physical system governed by the wave equation. Yamaha licensed the waveguide patents from Stanford in 1989 (Sondius package), resulting in the VL1 (1994), the first commercial physical modeling synthesizer. All original patents expired ~2010.

6. Hardware & Software

1978-83

Digital Chip

Karplus & Strong · custom VLSI · Intel 8080A · TI TMS9900

Custom VLSI chip reportedly supporting up to 16 polyphonic voices on early 8-bit microprocessors. The two-point average requires no multiplications — the algorithm was designed for minimal computation. The patent was licensed to a company that folded before shipping any product: the synth that never was.

[VLSI] [Intel 8080A] [TI TMS9900] [16 voices]

1986-1992

NeXT Music Kit

Jaffe & Smith · NeXT Cube · Motorola DSP56001 · Objective-C

First demonstrated October 12, 1988 at Davies Symphony Hall — the presentation ended with a live duet between the Cube and an SF Symphony violinist. First architecture to unify Music-N and MIDI paradigms. Released as open-source to CCRMA in 1992.

[NeXT Cube] [Motorola DSP56001] [Objective-C] [open-source 1992]

1994

Yamaha VL1

Physical modeling synthesizer · 2-voice polyphony · Sondius patent

First commercial synthesizer based on digital waveguide synthesis — direct descendant of KS. Extraordinarily expressive: breath, embouchure, and technique shaped the sound in real time. Only 2-voice polyphony because computation per voice was immense. A cult object.

[Yamaha] [Sondius] [waveguide synthesis] [2-voice]

2015

Mutable Instruments Rings

Eurorack module · 3 resonator models · V/oct + gate · open-source firmware

Brought physical string and modal resonator synthesis to Eurorack. Three models: KS-style, sympathetic strings, modal resonator. One of the most celebrated Eurorack modules ever made. Open-source firmware continued by the community.

[Eurorack] [Mutable Instruments] [open-source]

2025

Strymon Superkar+

Eurorack · 32-voice · Solo + Chord channels · 10HP

One of the most advanced dedicated hardware KS instruments currently available. 32 voices, two independent channels (Solo + Chord, each up to 16 voices), MIDI polyphony, CV/gate. Development led by Pete Celi (Strymon co-founder), from the Karplus mode in the Starlab reverb module.

7. Preset Reference

■ Historical Presets

Each preset is calibrated to a documented historical moment or instrument.

Digitar '78	Intel 8080A, Digitar VLSI chip. BitCrush + Decimation to simulate 8-bit output. Short burst, fast decay, zero body.
Acrobat '81	May All Your Children Be Acrobats, Jaffe 1981. First musical KS use. 8 guitars + mezzo-soprano. Warm nylon ensemble, ROOM body.
Piccolo Mandolin	Silicon Valley Breakdown — smallest imaginary instrument. Ultra-bright, NANO body, high pick position.
Plucked Gate	SVB — "immense bass plucked Golden Gate Bridge." Deep detune (-19st), TITAN body. Infra-bass.
Quadrastance	SVB — 4 ensembles in quadraphonic space. CATHEDRAL body, high Ensemble/stereo spread.
Transfer Function	Julius O. Smith recognizes KS = transfer function of ideal string. Material=0.5 exactly, PLATE body, pristine.
Telegram '84	Jaffe 1984. String quartet + tape. SPRING body, vibrato, warm cello character.
Grass '87	Jaffe 1987. SSA chorus + tape. VOID body, sine excitation, high vibrato — almost vocal.
WWW-89	NeXT Cube 1989. Jaffe & Smith for Jobs; Berners-Lee at CERN. Clean, precise, slightly cold. ROOM body.
Jobs Demo '88	October 12, 1988 — Davies Symphony Hall. Music Kit live duet with SF Symphony violinist. PLATE body, vibrato.
Venice Biennale	SVB world premiere 1983. CATHEDRAL body, maximum space — the body is the palazzo.

■ Conceptual Presets

King Strong	Tribute to Alex Strong — inventor. Material=0.5 exactly, bodyMix=0, no body. The algorithm as born.
Empire Str(ing)	A string 443m tall. TITAN body, detune -36st, very long decay. One pluck = a sound event.
Ghost Patent	The patent that went bankrupt before shipping. Slightly unstable detune, mild BitCrush — almost works.
Dulcimer de Leche	Dulcimer soaked in condensed milk. Soft material, SPRING body, slow caramel decay.
Commuted Body	Smith's commuted synthesis in practice. SPRING body, bodyMix=0.78 — the body IS the instrument.
Inf. Bluegrass	Strong was a bluegrass musician. Infinite banjo — ultra-percussive attack, bright ROOM body.
Wave Equation	Karplus & Strong tried to prove their algo solved the wave equation. Smith demonstrated that the KS transfer function corresponded to an idealized vibrating string model. That moment of recognition.
Shazam String	Smith co-invented Shazam's audio fingerprinting. Crystalline, unmistakable attack. ROOM body.
Sandburg Reed	Both Acrobats and Grass are Sandburg settings. Reedy, vocal, sine excitation, SPRING body.

CCRMA Room	The Stanford room where all of this was born. ROOM body, contemplative.
Public Domain '93	April 30, 1993 — WWW and Music Kit both open-sourced. VOID body, a string released into the commons.
Objective-C String	WorldWideWeb.app and the Music Kit share a mother tongue: Objective-C on NeXTSTEP. Structured, precise.

8. Vibe Coding & Philosophy

Strong+ was built through an extended dialogue between **Yuwa** — musician, composer, passionate amateur of synthesis and ambient textures, creator of the **Positive Lofi** YouTube channel — and Claude, an AI made by Anthropic. Yuwa is not a programmer. He is someone who loves sound deeply, who plays keyboards and dreams of modular synthesizers, who finds the EKS fascinating not as an engineer but as a musician who wants to understand what he hears. That question — what is actually happening inside this sound? — drove the entire conversation.

The NeXT Parallel

There is a remarkable symmetry between what happened on the NeXT Cube in 1988-1993 and what is happening now. Jaffe and Smith built the Music Kit on NeXT hardware — the most advanced workstation of its era. Berners-Lee built the first web browser on the same machine, in the same language (Objective-C), in the same years. Both were acts of enormous creative leverage: small teams, extraordinary tools, world-changing output. Both were released as open source — the Music Kit to CCRMA in 1992, the WWW to the public domain on April 30, 1993. The act of making and then releasing freely was not incidental. It was the whole point.

Strong+ inherits this lineage directly. The original patents have expired and the algorithm is freely implementable. The Music Kit was open-sourced. The web made distribution free. And vibe coding makes building radically accessible. The same compression that took KS from Stanford mainframes to browser JavaScript has now reached the act of creation itself.

What Is Authorship When the Tool Understands?

The architecture, preset values, historical research, and musical philosophy all emerged from dialogue. Yuwa brought the curiosity, the musical ear, the corrections, the direction, the insistence on accuracy. Claude brought the DSP implementation, the research, the code. Neither could have built this alone. The knowledge is distributed across the conversation in a way that doesn't belong to either party — exactly as in 1983, the knowledge of why KS worked was distributed between Strong (who discovered it), Karplus (who analyzed it), Jaffe (who musicalized it), and Smith (who proved it).

This is not a weakness. It is what collaborative making has always looked like, when we're honest about it.

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